Firebase setup: Analytics, Crashlytics, push (FCM)

The code is finished. Gameplay calls only two autoloads; each talks to a native plugin.

Service	Autoload	Native plugin
Analytics + Crashlytics (+ NDK)	`Analytics` (`src/autoload/analytics.gd`)	our **CastleFirebase** (`addons/castle_firebase`, Kotlin source `tools/android_plugins/firebase`)
Push (Firebase Cloud Messaging)	`Notifications` (`src/autoload/notifications_push.gd`)	**GodotX Firebase 3.1.0** Core + Messaging (`addons/godotx_firebase`, `android/firebase_core`, `android/firebase_messaging`, MIT)

Why two plugins: GodotX Firebase's Crashlytics module has no non-fatal error call (only `log_message`, custom keys and a test `crash()`). CastleFirebase sends every GDScript error to Crashlytics with its backtrace (`recordError`), which GodotX can't do. GodotX Messaging is a complete, maintained FCM module, so we use it instead of writing our own. Both use the same Firebase SDK versions (common 22.1, analytics 23.2, crashlytics 20.1, messaging 25.1), so the APK carries one consistent set. GodotX's own Analytics and Crashlytics modules stay **off** in the export preset.

Firebase switches on **only when `google-services.json` exists in the project root**. The GodotX export plugin copies it into `android/build/` at export time, where the Google services Gradle plugin reads it. Without the file builds work and send nothing. The file is git-ignored and is excluded from the packed game files.

Steps for the owner

1. https://console.firebase.google.com → **Add project** → "Stone Storm". Enable Google Analytics when asked. (Done: project `castle-smashers-a82d6`.)
2. **Add app** → **Android**:

Field	Value
Android package name	`com.castlesmashers.game` (exactly)
App nickname	Stone Storm
Debug signing certificate SHA-1	Optional (only needed for Google sign-in)

1. **Download `google-services.json`** and put it in the project root, next to `project.godot`. (Done: the file is there and its package name matches.)
2. **"Add Firebase SDK" step of the wizard: skip it.** Godot's Android template uses Groovy `build.gradle`, not the Kotlin DSL the wizard shows, and the plugins already add the SDKs. `python3 tools/patch_android_template.py` adds the two Gradle plugins (`google-services` 4.4.4, `firebase-crashlytics-gradle` 3.0.6, the versions GodotX is built and tested with). Run it again whenever you reinstall the Android build template. It is safe to run twice. Don't add the BoM line from the wizard: it would pull a second, newer set of Firebase libraries next to the plugins' set.
3. Export as usual (Project → Export → Android). The preset already has Use Gradle Build on and, under **Firebase**: Core on, Messaging on, Analytics off, Crashlytics off, config file `res://google-services.json`.
4. Firebase console → **Crashlytics** → "Enable Crashlytics". Start the game once on the phone.
5. Firebase console → **Messaging** → "Create your first campaign" → **Firebase Notification messages** → title + text → target **Topic** `all_players` (or `news`) → Review → Publish. Push the game to the background first: Android shows the notification itself.

6. Release builds: add the Play app-signing SHA-1 and SHA-256 (Play Console → Setup → App signing) in Firebase → Project settings → your app → **Add fingerprint**.
7. Data safety form: see below (FCM adds "Device or other IDs" for push delivery).

Checking it works

- **DebugView**: on a USB phone run `adb shell setprop debug.firebase.analytics.app com.castlesmashers.game`, open the game, and watch Firebase → Analytics → DebugView. Turn it off with `adb shell setprop debug.firebase.analytics.app .none.`
- Normal events show in the Analytics dashboard after up to 24 hours.
- **Crashlytics** reports show after the app is restarted following a crash.
- The player can turn both off in Settings → Analytics (`Analytics.set_enabled`).
- **Push**: `adb logcat -s GodotxFCMService GodotxFirebaseMessaging godot` shows the topic subscription a few seconds after start. Send a campaign to `all_players` with the game in the background; the notification uses the white castle icon. Turning notifications off in Settings unsubscribes the phone from every topic.

What is sent

Only whitelisted events and parameters (`Analytics.EVENTS`). Anything else is dropped. No names, emails, device IDs, purchase tokens or free text are ever logged.

Event	Parameters
`game_start`	`session_number`, `days_since_install`, `build`
`tutorial_complete`	`level_id`
`level_start`	`level_id`, `attempt`, `weapon`, `heroes`, `mode`
`level_complete`	`level_id`, `attempt`, `stars`, `score`, `duration_s`, `weapon`, `shots_used`, `used_hellfire`
`level_failed`	`level_id`, `attempt`, `reason`, `duration_s`, `weapon`, `shots_used`
`level_retry`	`level_id`
`level_quit`	`level_id`, `attempt`, `duration_s`, `shots_used`, `how` (map / restart from the pause menu)
`tutorial_begin`	`level_id`
`purchase_start`	`product_id`
`rewarded_ad_failed`	`placement`
`review_requested`	`level_id`
`update_prompted`	`mode` (flexible / immediate), `version_code`, `staleness_days`
`rewarded_ad_watched`	`placement` (spin / extra_rocky / double_coins), `reward_amount`, `level_id`
`interstitial_shown`	`level_id`
`purchase_success` / `purchase_failed`	`product_id`, `reward_amount` / `reason`
`crate_opened`	`product_id`, `reward_type`, `reward_amount`
`hellfire_used`	`level_id`, `left`
`upgrade_bought` / `engine_forged` / `skin_bought`	`weapon`, `track`, `level`, `cost` / `weapon`, `cost` / `skin_id`, `cost`
`spin_used`	`paid_with_ad`, `reward_type`, `reward_amount`

Event	Parameters
`notification_opened`	`kind`
`session_end`	`duration_s`, `levels_played`
`promo_shown` / `promo_tap`	`card` (noads / addons / skins / crates / upgrades / engines / missions)

Useful reports to build in Analytics → Explore: a funnel `level_start → level_complete` per `level_id` (finds hard levels), `attempt` at completion, and `rewarded_ad_watched` by placement.

Crashlytics: what each report contains

Need	How
Device model, Android version, RAM, orientation	Collected by Crashlytics automatically
Game version	Crashlytics app version + custom key `build`
Frequency	Crashlytics groups issues and counts crashes and affected users
Stack traces	Java/Kotlin automatically; native engine crashes via Crashlytics NDK; GDScript errors are sent as non-fatal issues with the GDScript backtrace (`Analytics.ErrorForwarder`, max 10 distinct per session)
Was it during physics, loading or an ad?	Custom key `phase`: `loading` (scene build), `aiming`, `physics` (after a shot), `ad_rewarded`, `ad_interstitial`; plus keys `scene`, `level`, `weapon`, `projectile`
What happened just before	Breadcrumb log: shots (`shot:<hero>:<weapon>`), rewarded placements

Data safety form (Play Console → App content → Data safety)

Answer **Yes, the app collects data**; it is encrypted in transit; users can't request deletion through the app (they can by email; see privacy policy).

Data type	Collected	Shared	Purpose	Optional?
App activity → App interactions	Yes	No	Analytics	Yes (Settings → Analytics)
App info and performance → Crash logs, Diagnostics	Yes	No	App functionality, Analytics	Yes
Device or other IDs (advertising ID, Firebase installation ID, FCM token)	Yes	Yes (AdMob)	Advertising, Analytics, Fraud prevention, App functionality (push)	No (AdMob)
Financial info → Purchase history	Yes	No	App functionality	No
Approximate location (from IP, AdMob)	Yes	Yes	Advertising	No